📂 Arduino Sonar Airplane Tracking System — Design Document

1. Project Summary

The Arduino Sonar Airplane Tracking System is a small electronic system that scans an area and detects objects using sound waves. It uses an ultrasonic sensor attached to a servo motor that rotates to scan from side to side, similar to how real radar systems track airplanes.

This project matters because it demonstrates how tracking technology works in aviation and engineering. It helps students understand how sensors, motors, and microcontrollers work together to detect objects and measure distance.

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2. Problem Statement

In aviation and transportation, tracking moving objects like airplanes is important for safety and navigation. However, real radar systems are expensive and complex.

This project solves the problem by creating a **small and simple model** that demonstrates how object detection and tracking works using low-cost electronics.

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3. Use Case

A possible use case is for students learning about robotics and electronics. They can use the system to scan an area and detect objects, helping them understand how real airplane tracking systems work.

The user will place the device in front of an open area. The servo motor will rotate the sensor and detect objects in different directions, sending distance information to the computer.

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4. Component Details & Behavior

Component	Description
Ultrasonic Sensor (HC-SR04)	Sends sound waves and measures the time it takes for the sound to bounce back to detect object distance.
Servo Motor	Rotates the ultrasonic sensor from left to right so the system can scan the surrounding area.
Arduino Uno	Controls the sensor and servo motor, processes the distance data, and sends the information to the computer.
Breadboard	Allows easy connection of electronic components without soldering.
Jumper Wires	Used to connect the Arduino, sensor, and servo motor together.
Computer	Runs the program and displays the distance readings from the sensor.

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5. How It Works (Control Logic Summary)

1. **Scanning**
   The servo motor moves the ultrasonic sensor from one side to the other.

2. **Detection**
   The ultrasonic sensor sends sound waves and measures the time it takes for the echo to return.

3. **Calculation**
   The Arduino calculates the distance between the sensor and the detected object.

4. **Data Output**
   The Arduino sends the distance information to the computer.

5. **Display**
   The computer displays the detected objects and distances on the serial monitor or visual radar display.

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6. Prototype Interactions (3+ Events)

1. Servo Rotation Event

When the program starts, the Arduino sends signals to the servo motor to rotate between angles.

Interaction:
Servo Motor → Arduino Control Signal → Sensor Position

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2. Object Detection Event

When the ultrasonic sensor detects an object within range, it measures the distance using echo timing.

Interaction:
Sensor → Arduino → Distance Calculation

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3. Data Display Event

After calculating the distance, the Arduino sends the data to the computer so it can be displayed.

Interaction:
Arduino → Computer → Serial Monitor Display

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7. Diagrams

Two diagrams will be included to explain the project:

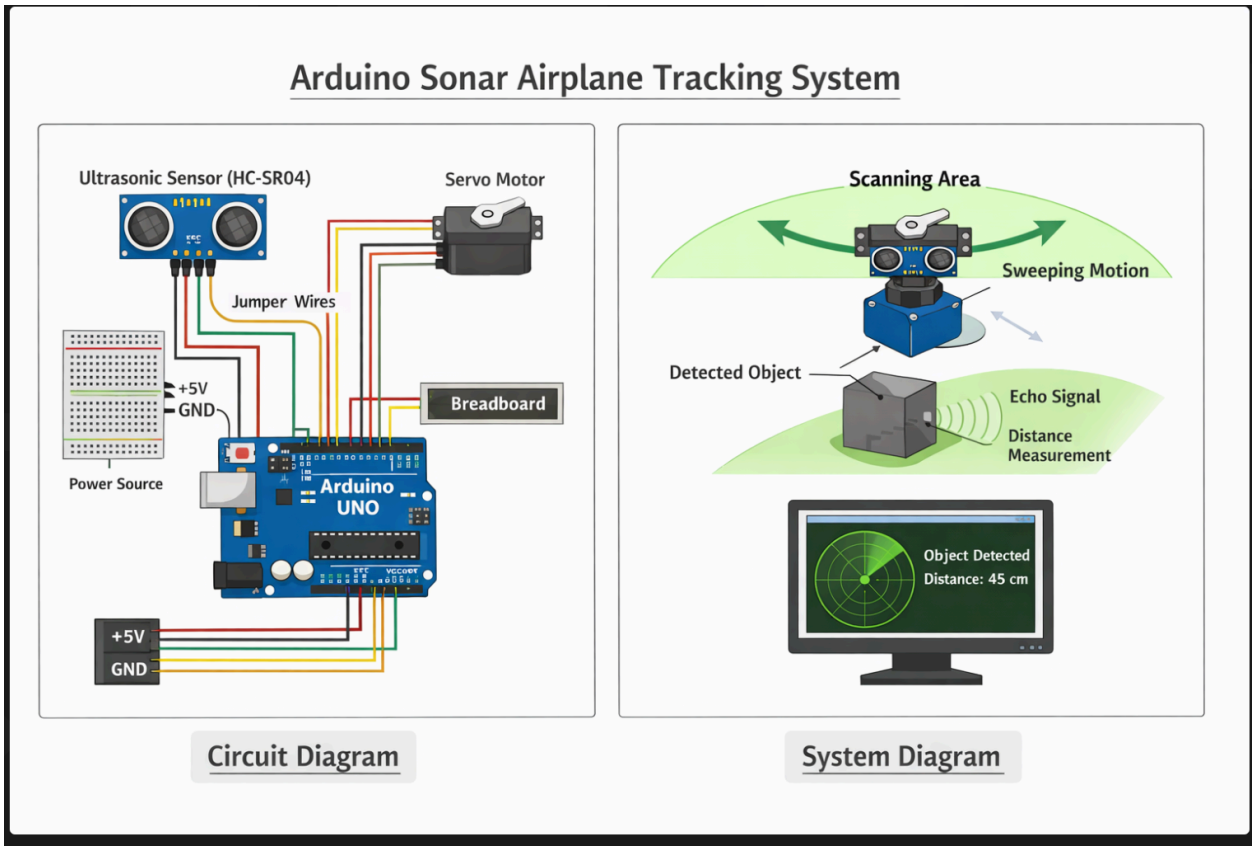
Version 1 – Circuit Diagram

- Shows how the ultrasonic sensor connects to the Arduino
- Shows servo motor wiring

- Includes power and ground connections

Version 2 – System Diagram

- Shows how the sensor scans the environment
- Demonstrates how objects are detected during rotation



```
          +-------------------+
          |   Ultrasonic      |
          |   Sensor          |
          |  (HC-SR04)        |
          +---------+---------+
                    |
                    | Distance Data
                    v
            +-------------+
            |   Arduino   |
            |     Uno     |
            +------+------+
                   |
       +-----------+-----------+
       |                       |
       v                       v
+-----------+           +-------------+
| Servo     |           | Computer /  |
| Motor     |           | Serial      |
| (Scanner) |           | Monitor     |
+-----------+           +-------------+
       |
       v
Rotates Sensor
to Scan Area
```

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8. Tech Stack and Tools

Hardware

- Arduino Uno
- Ultrasonic Sensor (HC-SR04)
- Servo Motor
- Breadboard
- Jumper Wires

Software

- Arduino IDE
- Processing

Programming Languages

- Arduino C/C++
- Processing (Java-based)

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9. Algorithm (High-Level)

1. Start the program on the Arduino.
2. Move the servo motor to the starting angle.
3. Trigger the ultrasonic sensor to send sound waves.
4. Measure the time it takes for the echo to return.
5. Calculate the distance to the object.
6. Send the distance data to the computer.
7. Move the servo to the next angle.
8. Repeat the scanning process.

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10. Flowchart

Start
↓
Initialize Arduino System
↓
Move Servo Motor
↓
Trigger Ultrasonic Sensor
↓
Measure Echo Time
↓
Calculate Distance
↓
Send Data to Computer
↓
Display Results
↓
Repeat Scan

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11. Timeline (2.5 Months)

Phase 1 – Research & Planning (2 Weeks)

- Research sonar and object detection
- Watch tutorials and plan wiring
- Gather all components

Phase 2 – Hardware Assembly (3 Weeks)

- Connect Arduino, sensor, and servo
- Build the circuit on a breadboard
- Test basic sensor readings

Phase 3 – Programming (3 Weeks)

- Write Arduino code
- Program servo scanning movement
- Send sensor data to computer

Phase 4 – Testing & Final Adjustments (2 Weeks)

- Test detection accuracy
- Fix coding issues
- Finalize project documentation

Total Time ≈ **2.5 Months**

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12. Risk Mitigation

Risk:
The ultrasonic sensor might give incorrect distance readings due to interference or incorrect wiring.

Mitigation Plan:
Test the sensor multiple times and adjust the code to filter incorrect values and improve accuracy.

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13. Evaluation Criteria

The project will be considered successful if:

1. The servo motor rotates smoothly while scanning.
2. The ultrasonic sensor correctly detects objects and measures their distance.
3. The system sends accurate data to the computer for display.

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14. Future Considerations

Maintenance

Check wiring connections regularly to ensure the sensor and servo motor work properly.

Future Improvements

In the future, the project could include a **full graphical radar interface** that displays detected objects in real time on a screen.